

Parul University

Vadodara, Gujarat, India

Bachelor of Technology - Computer Science and Engineering; GPA: 7.75/10

June 2022 - Expected June 2026

Relevant Coursework:

Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Object-Oriented Programming (Java)

SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C, SQL
- **AI / Machine Learning:** PyTorch, TensorFlow, YOLO, NLP, LLMs, LangChain, RAG
- **Backend & Systems:** FastAPI, Node.js, REST APIs, Distributed Systems, Redis
- **Databases:** PostgreSQL, MongoDB, SQLite, Vector Databases
- **Web & Tools:** React.js, Docker, Git, Linux, CI/CD
- **Soft Skills:** Ownership, Technical Leadership, Systems Thinking, Execution Under Pressure

EXPERIENCE

Upsquare Technologies

India

Artificial Intelligence Intern

Jan 2026 - Present

• **Computer Vision Pipelines:** Developed and debugged video analytics pipelines for real-world CCTV and transportation environments.

• **Model Training and Evaluation:** Prepared datasets and conducted YOLO-based fine-tuning with evaluation across occlusion, lighting, and camera-angle variations.

• **System Integration:** Connected AI inference workflows with backend services and a web-based upload-to-results pipeline.

PROJECTS

AID: AI Identity Passport

AI Infrastructure Architect

AI Infrastructure Project (In Development)

2025 – Present

• **Identity and Trust Layer:** Designed a decentralized identity framework composed of 3 core components (identity issuance, verification, and lineage binding) to bind agent identity across multi-step workflows.

• **Security Enforcement:** Defined identity-first execution policies applied to 100% of agent actions in the prototype design, enabling per-execution attribution and traceability across chained agent operations.

Octopus: Autonomous Multi-Agent Orchestration System

AI Systems Architect

AI Systems Project

2025

• **System Architecture:** Designed an orchestration layer coordinating workflows across 5 integrated production tools and 2 execution phases (planning and execution), replacing manual handoffs present in baseline pipelines.

• **Memory and Reasoning:** Introduced shared vector memory and structured reasoning pipelines supporting 10+ concurrent agent executions across multi-step tasks without context resets.

• **Automation Impact:** Observed a 35–40% decrease in end-to-end task completion time and a 2x increase in reusable workflows during repeated task executions using standardized agent templates.

Memron: Decentralized Multi-Agent Persistent Protocol

AI Engineer

Distributed Systems / Blockchain Project

2025

• **Persistent Agent Memory:** Engineered a decentralized memory layer spanning 2 storage layers (Avalanche for verification and IPFS for persistence) to retain agent state across multiple execution sessions.

• **Cross-Agent Communication:** Developed synchronization protocols enabling state propagation between 5–8 concurrently running agents with sub-second consistency delays in controlled test environments.

• **Integrity and Performance:** Measured a 45–50% reduction in context retrieval latency and a 35–40% drop in lost or overwritten memory events compared to non-deterministic baseline reads.

VisionX: AI-Driven Developer Collaboration Platform

Full Stack Engineer

Full Stack Systems Project

2023 – 2024

• **Platform Development:** Built a GitHub-integrated collaboration platform consisting of 6 major modules (auth, PR validation, analytics, incentives, workflows, dashboards) used across multiple active repositories.

• **Impact:** Recorded a 50–60% increase in contributor participation and a 40–45% reduction in manual PR review effort during pilot usage across repeated contribution cycles.

PUBLICATIONS

- **The Cost of Coordination: Scheduling and Consistency in Multi-Agent AI Systems (2025):** Lead author. Designed and validated the Token-Latency-Cost Optimization (TLCO) scheduling framework alongside the HCP-A causal consistency protocol to mitigate coordination overhead and stale state in multi-agent systems. **Results:** Achieved 45% improvement in scheduling efficiency and 40% reduction in stale reads under high-concurrency simulations.